EC24101 BASIC ELECTRONICS

Module 2: Bipolar Junction Transistors (BJT)

Dr. Jyoti Sharma

BIT Mesra, Jaipur Campus

October 14, 2024

Overview

1. Introduction
2. Basic operation of PNP and NPN Transistors
3. Input output characteristics of CE, CB, and CC configurations
4. Transistor biasing: operating point
5. Fixed bias
6. Emitter bias
7. Voltage divider bias
8. Stability factor
9. Small signal analysis (h-parameters model) of CE configuration

Transistor Construction

- The transistor is a three-layer semiconductor device consisting of either two n- and one p-type layers of material or two p- and one n-type layers of material.
- The former is called an npn transistor, while the latter is called a pnp transistor.
- Both are shown in Fig. 1 with the proper dc biasing.
- The dc biasing is necessary to establish the proper region of operation for ac amplification.
- The emitter layer is heavily doped, the base lightly doped, and the collector only lightly doped.
- The outer layers have widths much greater than the sandwiched p- or n-type material.
- The doping of the sandwiched layer is also considerably less than that of the outer layers. This lower doping level decreases the conductivity (increases the resistance) of this material by limiting the number of "free" carriers.

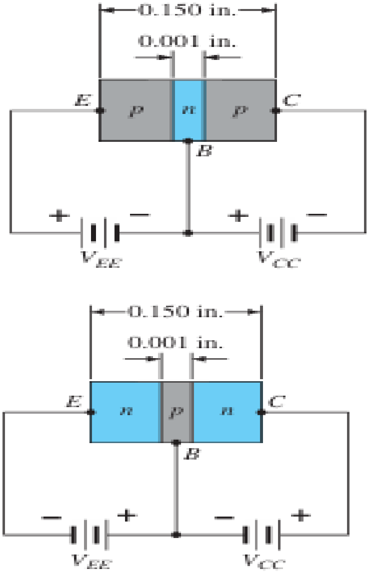


Figure 1: Types of Transistors (a) pnp (b) npn

- For the biasing shown in Fig. 1 the terminals have been indicated by the capital letters E for emitter, C for collector, and B for base.
- The abbreviation BJT, from bipolar junction transistor, is often applied to this three terminal device.
- The term bipolar reflects the fact that holes and electrons participate in the injection process into the oppositely polarized material.
- If only one carrier is employed (electron or hole), it is considered a unipolar device.

Transistor Operation

- The basic operation of the transistor is described here using the pnp transistor of Fig. 1(a).
- The operation of the npn transistor is exactly the same if the roles played by the electron and hole are interchanged.
- In Fig. 2, the pnp transistor has been redrawn without the base-to-collector bias.
- Note the similarities between this situation and that of the forward-biased diode in Module 1.
- The depletion region has been reduced in width due to the applied bias, resulting in a heavy flow of majority carriers from the p- to the n-type material.

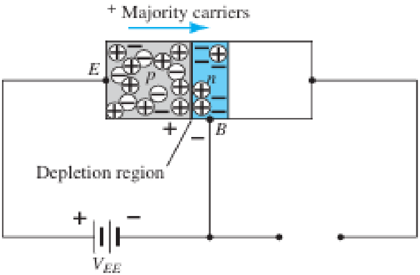


Figure 2: Biasing a Transistor- Forward Bias

- Let us now remove the base-to-emitter bias of the pnp transistor of Fig.1(a) as shown in Fig.3.
- Consider the similarities between this situation and that of the reverse-biased diode. Recall that the flow of majority carriers is zero, resulting in only a minority-carrier flow, as indicated in Fig.3.
- In summary, therefore: **One p–n junction of a transistor is reverse-biased, whereas the other is forward-biased.**

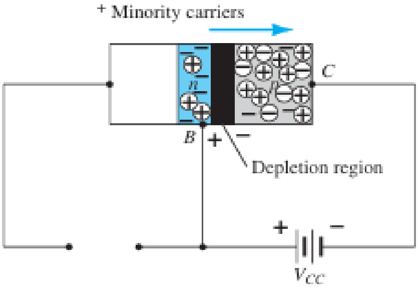


Figure 3: Biasing a Transistor- Reverse Bias

- In Fig.4 both biasing potentials have been applied to a pnp transistor, with the resulting majority- and minority-carrier flows indicated.
- Note the widths of the depletion regions, indicating clearly which junction is forward-biased and which is reverse-biased.
- As indicated here, a large number of majority carriers will diffuse across the forward biased p–n junction into the n -type material.

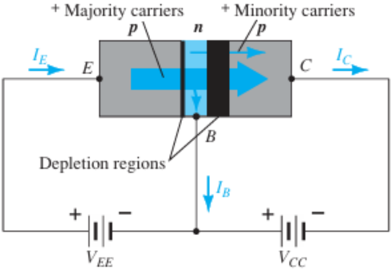


Figure 4: Majority and minority carrier flow of a pnp transistor

- The question then is whether these carriers will contribute directly to the base current I_B or pass directly into the p-type material.
- Since the sandwiched n -type material is very thin and has a low conductivity, a very small number of these carriers will take this path of high resistance to the base terminal.
- The magnitude of the base current is typically on the order of microamperes, as compared to milliamperes for the emitter and collector currents.
- The larger number of these majority carriers will diffuse across the reverse-biased junction into the p -type material connected to the collector terminal as indicated in Fig.4.

- The reason for the relative ease with which the majority carriers can cross the reverse-biased junction is easily understood if we consider that for the reverse-biased diode the injected majority carriers will appear as minority carriers in the n -type material.
- In other words, there has been an injection of minority carriers into the n -type base region material.
- Combining this with the fact that all the minority carriers in the depletion region will cross the reverse-biased junction of a diode accounts for the flow indicated figure.

- Applying Kirchhoff's current law to the transistor of Fig.4 as if it were a single node, we obtain
  $I_E = I_C + I_B$
- and find that the emitter current is the sum of the collector and base currents. The collector current, however, comprises two components—the majority and the minority carriers as indicated in Fig.4.
- The minority-current component is called the leakage current and is given the symbol I_{CO} (I_C current with emitter terminal Open).
- The collector current, therefore, is determined in total by
  $I_C = I_{Cmajority} + I_{Cminority}$

- For general-purpose transistors, I_C is measured in milliamperes and I_{CO} is measured in microamperes or nanoamperes.
- I_{CO}, like I_s for a reverse-biased diode, is temperature sensitive and must be examined carefully when applications of wide temperature ranges are considered.
- It can severely affect the stability of a system at high temperature if not considered properly.

- The notation and symbols used in conjunction with the transistor are indicated in Fig.5 for the common-base configuration with pnp and npn transistors.

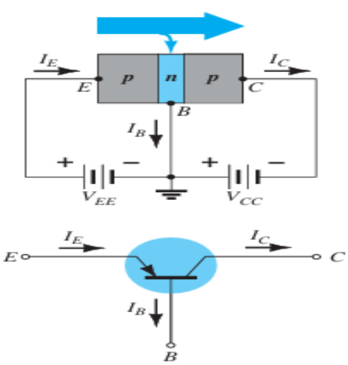


Figure 5: Notation and symbols used with the common-base configuration: (a) pnp transistor; (b) npn transistor

- The common-base terminology is derived from the fact that the base is common to both the input and output sides of the configuration.
- In addition, the base is usually the terminal closest to, or at, ground potential.
- Throughout the lecture, all current directions will refer to conventional (hole) flow rather than electron flow.
- The result is that the arrows in all electronic symbols have a direction defined by this convention.
- Recall that the arrow in the diode symbol defined the direction of conduction for conventional current.
- For the transistor: **The arrow in the graphic symbol defines the direction of emitter current (conventional flow) through the device.**

- The applied biasing (voltage sources) are such as to establish current in the direction indicated for each branch.
- That is, compare the direction of I_E to the polarity of V_{EE} for each configuration and the direction of I_C to the polarity of V_{CC}.
- To fully describe the behavior of a three-terminal device such as the common-base amplifiers of Fig.5 requires two sets of characteristics—one for the driving point or input parameters and the other for the output side.

Input Characteristics

- The input set for the common-base amplifier as shown in Fig.6 relates an input current (I_E) to an input voltage (V_{BE}) for various levels of output voltage (V_{CB}).
- The input characteristics of Fig.6 reveal that for fixed values of collector voltage (V_{CB}), as the base-to-emitter voltage increases, the emitter current increases in a manner that closely resembles the diode characteristics.

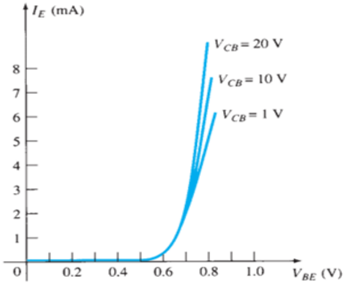


Figure 6: Input or driving point characteristics

Output Characteristics

- The output set relates an output current (I_C) to an output voltage (V_{CB}) for various levels of input current (I_E) as shown in Fig. 7.
- There are three basic regions of interest: the active, cutoff, and saturation regions.

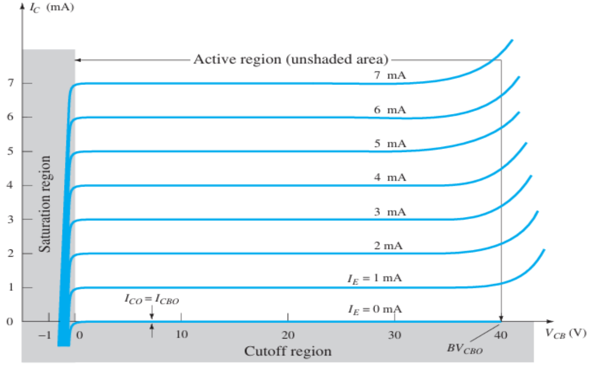


Figure 7: Output or collector characteristics

Output Characteristics

- The **active region** is the region normally employed for linear (undistorted) amplifiers.
- **In the active region the base–emitter junction is forward-biased, whereas the collector base junction is reverse-biased.**
- At the lower end of the active region the emitter current (I_E) is zero, and the collector current is simply that due to the reverse saturation current I_{CO} , as indicated in Fig.8.

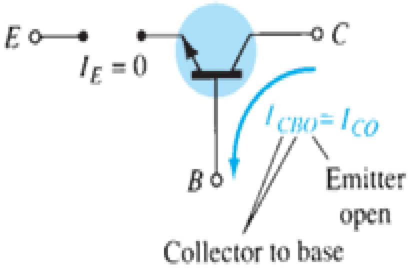


Figure 8: Output or collector characteristics

- The current I_{CO} is so small (microamperes) in magnitude compared to the vertical scale of I_C(milliamperes) that it appears on virtually the same horizontal line as I_C=0.
- The current I_{CO} is also enoted as I_{CBO}.
- In addition, keep in mind that I_{CBO}, like I_s, for the diode (both reverse leakage currents) is temperature sensitive.
- At higher temperatures the effect of $ICBO$ may become an important factor since it increases so rapidly with temperature.

- As the emitter current increases above zero, the collector current increases to a magnitude essentially equal to that of the emitter current as determined by the basic transistor-current relations.
- Note also the almost negligible effect of V_{CB} on the collector current for the active region.
- The curves clearly indicate that a first approximation to the relationship between I_E and I_C in the active region is given by $I_C \approxeq I_E$.
- The **cutoff region** is defined as that region where the collector current is 0 A.
- **In the cutoff region the base–emitter and collector–base junctions of a transistor are both reverse-biased.**

- The saturation region is defined as that region of the characteristics to the left of V_{CB}=0 V.
- The horizontal scale in this region was expanded to clearly show the dramatic change in characteristics in this region.
- Note the exponential increase in collector current as the voltage V_{CB} increases toward 0 V.
- **In the saturation region the base–emitter and collector–base junctions are forward-biased.**

Current Gain, Alpha (α)

- **DC Mode**: In the dc mode the levels of I_C and I_E due to the majority carriers are related by a quantity called alpha and defined by the following equation:

  $$\alpha_{dc} = \frac{I_C}{I_E}$$

  where I_C and I_E are the levels of current at the point of operation.
- Ideally $\alpha =\approx 1$, but for practical devices alpha typically extends from 0.90 to 0.998.
- Since α is defined solely for the majority carriers, $I_C = I_{Cmajority} + I_{Cminority} = \alpha I_E + I_{CBO}$
- For the output characteristics when I_E=0 mA, I_C is therefore equal to I_{CBO}, but the level of I_{CBO} is usually so small that it is virtually undetectable on the graph.
- In other words, when I_E=0 mA, I_C also appears to be 0 mA for the range of V_{CB} values.

Current Gain, Alpha (α)

- **AC Mode**: For ac situations where the point of operation moves on the characteristic curve, an ac alpha is defined by

$$\alpha_{ac} = \frac{\Delta I_C}{\Delta I_E}\Big|_{V_{CB}=constant}$$

- The ac alpha is called the common-base, short-circuit, amplification factor

Breakdown Region

- As the applied voltage V_{CB} increases there is a point where the curves take a dramatic upswing in the output characteristics.
- This is due primarily to an avalanche effect similar to that described for the diode, when the reverse-bias voltage reached the breakdown region.
- As the base-to-collector junction is reversed biased in the active region, but there is a point where too large a reverse-bias voltage will lead to the avalanche effect.
- The result is a large increase in current for small increases in the base-to-collector voltage.
- The largest permissible base-to-collector voltage is labeled BV_{CBO} as shown in the output characteristics. It is also referred to as $V_{(BR)CBO}$.
- The uppercase letter O is used to represent that the emitter leg is in the open state (not connected).

- It is called the common-emitter configuration because the emitter is common to both the input and output terminals.
- Two sets of characteristics are again necessary to describe fully the behavior of the common-emitter configuration: one for the input or base–emitter circuit and one for the output or collector–emitter circuit.

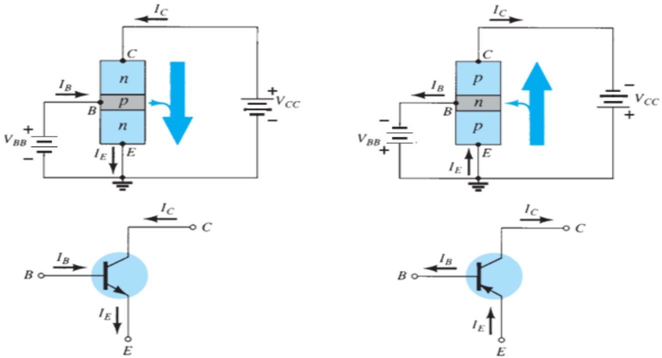


Figure 9: Common-emitter configuration: (a) npn transistor; (b) pnp transistor.

Common Emitter Configuration

- The current relations developed earlier for the common-base configuration are still applicable. That is, $I_E = I_C + I_B$ and $I_C = \alpha I_E$.
- The magnitude of I_B is in microamperes, compared to milliamperes of I_C.

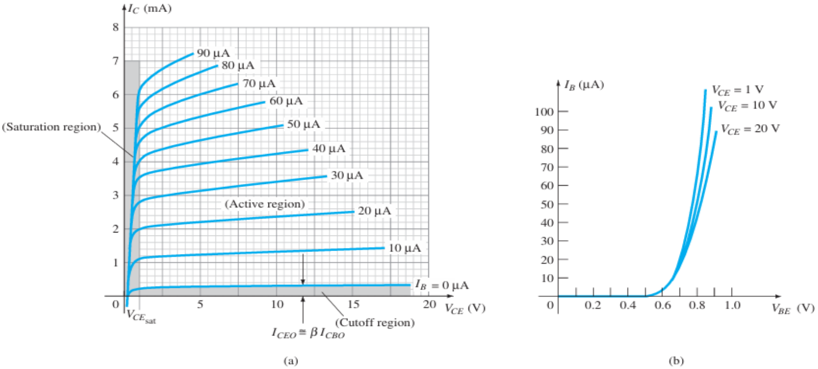


Figure 10: Characteristics of a silicon transistor in the common-emitter configuration: (a) collector characteristics; (b) base characteristics.

- Consider also that the curves of I_B are not as horizontal as those obtained for I_E in the common-base configuration, indicating that the collector-to-emitter voltage will influence the magnitude of the collector current.
- The active region for the common-emitter configuration is that portion of the upper-right quadrant that has the greatest linearity, that is, that region in which the curves for I_B are nearly straight and equally spaced.
- In Fig. 10(a) this region exists to the right of the vertical dashed line at $V_{CE_{sat}}$ and above the curve for I_B equal to zero.
- The region to the left of $V_{CE_{sat}}$ is called the saturation region.
- **In the active region of a common-emitter amplifier, the base–emitte junction is forward-biased, whereas the collector–base junction is reverse-biased.**

Common Emitter Configuration

- The active region of the common-emitter configuration can be employed for voltage, current, or power amplification.
- The cutoff region for the common-emitter configuration is not as well defined as for the common-base configuration.
- Note on the collector characteristics that I_C is not equal to zero when I_B is zero.
- For the common-base configuration, when the input current I_E was equal to zero, the collector current was equal only to the reverse saturation current I_{CO}.

$$I_C = \alpha I_E + I_{CBO}$$

$$I_C = \alpha(I_C + I_B) + I_{CBO}$$

$$I_C = \frac{\alpha I_B}{1-\alpha} + \frac{I_{CBO}}{1-\alpha}$$

when I_B=0 A, and $\alpha \approx 0.996$

$$I_C = \frac{\alpha(0A)}{1-0.996} + \frac{I_{CBO}}{1-0.996}$$

$$\frac{I_{CBO}}{0.004} = 250 I_{CBO}$$

If I_{CBO}=0 μA, the resulting collector current with I_B=0 A would be 250(1 μA)=0.25 mA, as reflected in the characteristics.

$$I_{CEO} = \left.\frac{I_{CBO}}{1-\alpha}\right|_{I_B=0\mu A}$$

- For linear (least distortion) amplification purposes, cutoff for the common-emitter configuration will be defined by $I_C = I_{CEO}$.
- The region below I_B=0 μA is to be avoided if an undistorted output signal is required.
- When employed as a switch in the logic circuitry of a computer, a transistor will have two points of operation of interest: one in the cutoff and one in the saturation region.

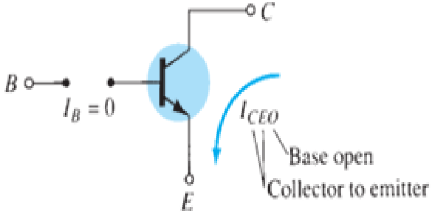


Figure 11: Circuit conditions related to I_{CEO}.

Beta (β)

- **DC Mode:** In the dc mode the levels of I_C and I_B are related by a quantity called beta and defined by the following equation:

  $$\beta_{dc} = \frac{I_C}{I_B}$$

  where I_C and I_B are determined at a particular operating point on the characteristics. For practical devices the level of β typically ranges from about 50 to over 400, with most in the midrange.
- **AC Mode:** For ac situations an ac beta is defined as follows:

  $$\beta_{ac} = \frac{\Delta I_C}{\Delta I_B}\bigg|_{V_{CE}=constant}$$

- The formal name for β_{ac} is common-emitter, forward-current, amplification factor.

- A relationship can be developed between β and α using the basic relationships introduced thus far.
- Using $\beta = \frac{I_C}{IB}$, we have $I_B = \frac{I_C}{\beta}$, and from $\alpha = \frac{I_C}{I_E}$ we have $I_E = \frac{I_C}{\alpha}$. Substituting into
  $I_E = I_C + I_B$
  $\frac{I_C}{\alpha} = I_C + \frac{I_C}{\beta}$
  and dividing both sides of the equationby I_C results in
  $\frac{1}{\alpha} = 1 + \frac{1}{\beta}$
  $\beta = \alpha\beta + \alpha = (\beta + 1)\alpha$
  $\alpha = \frac{\beta}{\beta+1}$
  $\beta = \frac{\alpha}{1-\alpha}$
  Also, $I_{CEO} = \frac{I_{CBO}}{1-\alpha}$
  As we have, $\frac{1}{1-\alpha} = \beta + 1$
  Thus, $I_{CEO} = (\beta + 1)I_{CBO}$
  $I_{CEO} \approx \beta I_{CBO}$
  $I_C = \beta I_B$
  $I_E = I_C + I_B = (\beta + 1)I_B$

Common Collector Configuration

- The common-collector configuration is used primarily for impedance-matching purposes since it has a high input impedance and low output impedance.

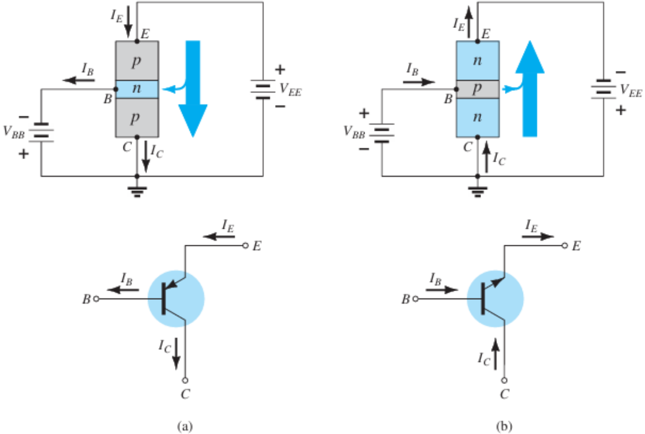


Figure 12: Notation and symbols used with the common-collector configuration:

Common Collector Configuration

- For all practical purposes, the output characteristics of the common-collector configuration are the same as for the common-emitter configuration.
- For the common-collector configuration the output characteristics are a plot of I_E versus V_{CE} for a range of values of I_B.
- The input current, therefore, is the same for both the common emitter and common-collector characteristics.
- The horizontal voltage axis for the common collector configuration is obtained by simply changing the sign of the collector-to-emitter voltage of the common-emitter characteristics.
- Finally, there is an almost unnoticeable change in the vertical scale of I_C of the common-emitter characteristics if I_C is replaced by I_E for the common-collector characteristics (since α=1).

Summary

- Semiconductor devices have the following advantages over vacuum tubes: They are (1) of smaller size , (2) more lightweight , (3) more rugged , and (4) more efficient. In addition, they have (1) no warm-up period , (2) no heater requirement , and (3) lower operating voltages.
- Transistors are three-terminal devices of three semiconductor layers having a base or center layer a great deal thinner than the other two layers. The outer two layers are both of either n - or p -type materials, with the sandwiched layer the opposite type.
- One p – n junction of a transistor is forward-biased , whereas the other is reverse biased.
- The dc emitter current is always the largest current of a transistor, whereas the base current is always the smallest . The emitter current is always the sum of the other two.
- The collector current is made up of two components : the majority component and the minority current (also called the leakage current).

Summary

- The arrow in the transistor symbol defines the direction of conventional current flow for the emitter current and thereby defines the direction for the other currents of the device.
- A three-terminal device needs two sets of characteristics to completely define its characteristics.
- In the active region of a transistor, the base–emitter junction is forward-biased, whereas the collector–base junction is reverse-biased.
- In the cutoff region the base–emitter and collector–base junctions of a transistor are both reverse-biased.
- In the saturation region the base–emitter and collector–base junctions are forward biased.
- On an average basis, as a first approximation, the base-to-emitter voltage of an operating transistor can be assumed to be 0.7 V.
- The quantity alpha (a) relates the collector and emitter currents and is always close to one.

Summary

- The impedance between terminals of a forward-biased junction is always relatively small, whereas the impedance between terminals of a reverse-biased junction is usually quite large.
- The arrow in the symbol of an npn transistor points out of the device (not pointing in), whereas the arrow points in to the center of the symbol for a pnp transistor (pointing in).
- For linear amplification purposes, cutoff for the common-emitter configuration will be defined by $I_C = I_{CEO}$.
- The quantity beta (b) provides an important relationship between the base and collector currents, and is usually between 50 and 400.
- The dc beta is defined by a simple ratio of dc currents at an operating point, whereas the ac beta is sensitive to the characteristics in the region of interest. For most applications, however, the two are considered equivalent as a first approximation.
- To ensure that a transistor is operating within its maximum power level rating, simply find the product of the collector-to-emitter voltage and the collector current, and compare it to the rated value.

Transistor Biasing

- The analysis or design of a transistor amplifier requires a knowledge of both the dc and the ac response of the system.
- it is assumed that the transistor is a magical device that can raise the level of the applied ac input without the assistance of an external energy source.
- In actuality, any increase in ac voltage, current, or power is the result of a transfer of energy from the applied dc supplies.
- The analysis or design of any electronic amplifier therefore has two components: a dc and an ac portion.
- Fortunately, the superposition theorem is applicable, and the investigation of the dc conditions can be totally separated from the ac response.
- However, during the design stage the choice of parameters for the required dc levels will affect the ac response, and vice versa.

Transistor Biasing

- The dc level of operation of a transistor is controlled by a number of factors, including the range of possible operating points on the device characteristics.
- Once the desired dc current and voltage levels have been defined, a network must be constructed that will establish the desired operating point.
- A number of these networks are analyzed in this module.
- Each design will also determine the stability of the system, that is, how sensitive the system is to temperature variations.

- The following important basic relationships for a transistor will be used for all the networks:

$$V_{BE} \approx 0.7V$$

$$I_E = (\beta + 1)I_B \approx I_C$$

$$I_C = \beta I_B$$

Operating Point

- The term biasing appearing in the title of this chapter is an all-inclusive term for the application of dc voltages to establish a fixed level of current and voltage.
- For transistor amplifiers the resulting dc current and voltage establish an operating point on the characteristics that define the region that will be employed for amplification of the applied signal.
- Because the operating point is a fixed point on the characteristics, it is also called the quiescent point (abbreviated Q -point).
- By definition, quiescent means quiet, still, inactive.

- Figure 13 shows a general output device characteristic with four operating points indicated.

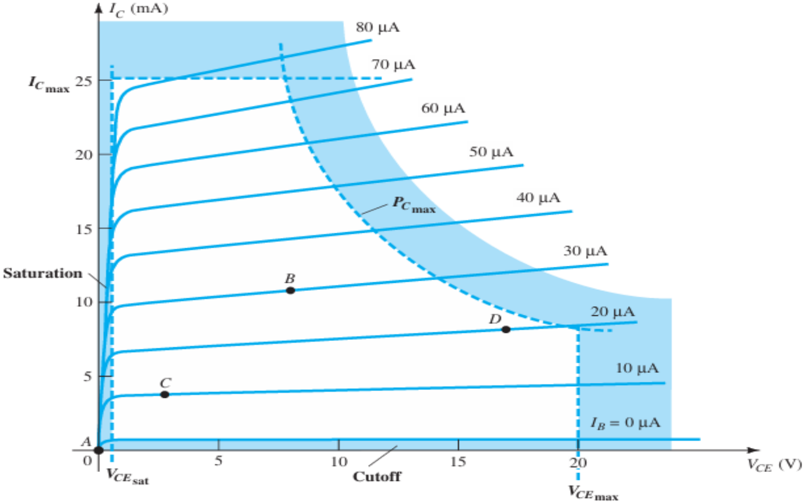


Figure 13: Various operating points within the limits of operation of a transistor.

- The biasing circuit can be designed to set the device operation at any of these points or others within the active region.
- The maximum ratings are indicated on the characteristics of Figure 13 by a horizontal line for the maximum collector current $I_{C_{max}}$ and a vertical line at the maximum collector-to-emitter voltage $V_{CE_{max}}$.
- The maximum power constraint is defined by the curve $P_{C max}$ in the same figure.
- At the lower end of the scales are the cutoff region, defined by $I_B le0mA$, and the saturation region, defined by $V_{CE} \leq V_{CE_{sat}}$.
- The BJT device could be biased to operate outside these maximum limits, but the result of such operation would be either a considerable shortening of the lifetime of the device or destruction of the device.

- If no bias were used, the device would initially be completely off, resulting in a Q point at A —namely, zero current through the device (and zero voltage across it).
- Because it is necessary to bias a device so that it can respond to the entire range of an input signal, point A would not be suitable.
- For point B, if a signal is applied to the circuit, the device will vary in current and voltage from the operating point, allowing the device to react to (and possibly amplify) both the positive and negative excursions of the input signal.
- If the input signal is properly chosen, the voltage and current of the device will vary, but not enough to drive the device into cutoff or saturation.
- Point C would allow some positive and negative variation of the output signal, but the peak-to-peak value would be limited by the proximity of V_{CE}= 0 V and I_C = 0 mA.
- Operating at point C also raises some concern about the nonlinearities introduced by the fact that the spacing between I_B curves is rapidly changing in this region.
- In general, it is preferable to operate where the gain of the device

- If no bias were used, the device would initially be completely off, resulting in a Q point at A —namely, zero current through the device (and zero voltage across it).
- Because it is necessary to bias a device so that it can respond to the entire range of an input signal, point A would not be suitable.
- For point B, if a signal is applied to the circuit, the device will vary in current and voltage from the operating point, allowing the device to react to (and possibly amplify) both the positive and negative excursions of the input signal.
- If the input signal is properly chosen, the voltage and current of the device will vary, but not enough to drive the device into cutoff or saturation.
- Point C would allow some positive and negative variation of the output signal, but the peak-to-peak value would be limited by the proximity of $V_{CE} = 0$ V and $I_C = 0$ mA.

- Operating at point C also raises some concern about the nonlinearities introduced by the fact that the spacing between I_B curves is rapidly changing in this region.
- In general, it is preferable to operate where the gain of the device is fairly constant (or linear) to ensure that the amplification over the entire swing of input signal is the same. Point B is a region of more linear spacing and therefore more linear operation.
- Point D sets the device operating point near the maximum voltage and power level. The output voltage swing in the positive direction is thus limited if the maximum voltage is not to be exceeded.
- Point B therefore seems the best operating point in terms of linear gain and largest possible voltage and current swing.

- Having selected and biased the BJT at a desired operating point, we must also take the effect of temperature into account.
- Temperature causes the device parameters such as the transistor current gain (β_{ac}) and the transistor leakage current (I_{CEO}) to change.
- Higher temperatures result in increased leakage currents in the device, thereby changing the operating condition set by the biasing network.
- The result is that the network design must also provide a degree of temperature stability so that temperature changes result in minimum changes in the operating point.
- This maintenance of the operating point can be specified by a stability factor S, which indicates the degree of change in operating point due to a temperature variation. A highly stable circuit is desirable.

- For the BJT to be biased in its linear or active operating region the following must be true:
  • The base–emitter junction must be forward-biased (p-region voltage more positive), with a resulting forward-bias voltage of about 0.6 V to 0.7 V.
  • The base–collector junction must be reverse-biased (n-region more positive), with the reverse-bias voltage being any value within the maximum limits of the device.

- Operation in the cutoff, saturation, and linear regions of the BJT characteristic are provided as follows:

  1. **Linear-region operation:**

  Base–emitter junction forward-biased
  Base–collector junction reverse-biased

  2. **Cutoff-region operation:**

  Base–emitter junction reverse-biased
  Base–collector junction reverse-biased

  3. **Saturation-region operation:**

  Base–emitter junction forward-biased
  Base–collector junction forward-biased

Fixed-Bias Configuration

- For the dc analysis the network can be isolated from the indicated ac levels by replacing the capacitors with an open-circuit equivalent because the reactance of a capacitor is a function of the applied frequency.
- For dc, f = 0 Hz, and $X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(0)C} = \infty\Omega$.
- In addition, the dc supply V_{CC} can be separated into two supplies.

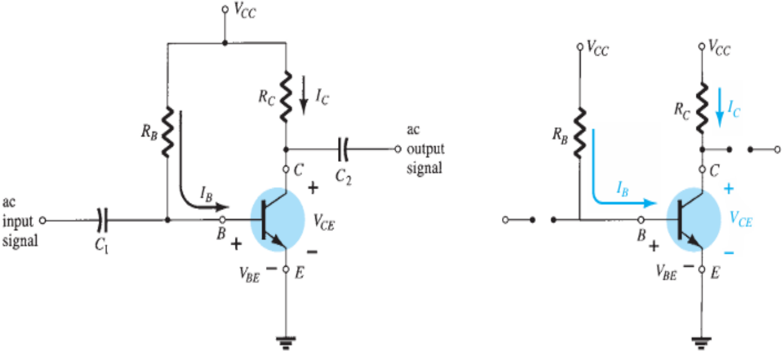


Figure 14: (a) Fixed-bias circuit (b) It's DC equivalent

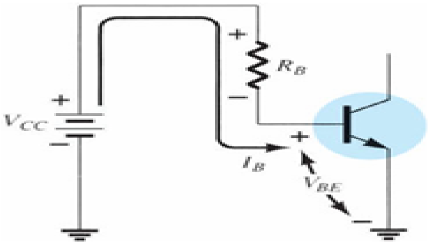


Figure 15: Base–emitter loop.

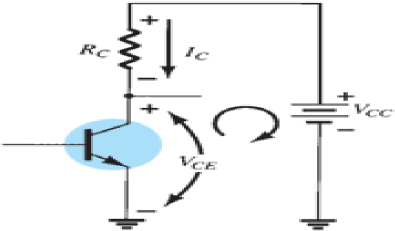


Figure 16: Collector–emitter loop.

Base–emitter loop

- Consider first the base–emitter circuit loop of Figure 15. Writing Kirchhoff's voltage equation in the clockwise direction for the loop, we obtain:

  $+V_{CC} - I_B R_B - V_{BE} = 0$

  $I_B = \frac{V_{CC} - V_{BE}}{R_B}$
- Since the supply voltage V_{CC} and the base–emitter voltage V_{BE} are constants, the selection of a base resistor R_B sets the level of base current for the operating point.

Collector–emitter loop

$I_C = \beta I_B$

Applying Kirchhoff's voltage law in the clockwise direction in Fig.16:

$V_{CE} + I_C R_C - V_{CC} = 0$

$V_{CE} = V_{CC} - I_C R_C$

$V_{CE} = V_C - V_E$

Since V_E=0 V, $V_{CE} = V_C$

Similarly, $V_{BE} = V_B$

Example

- Determine the following for the fixed-bias configuration:

(a) I_{B_Q} and I_{C_Q}
(b) $V_{CE\,Q}$
(c) V_B and V_C
(d) V_{BC}

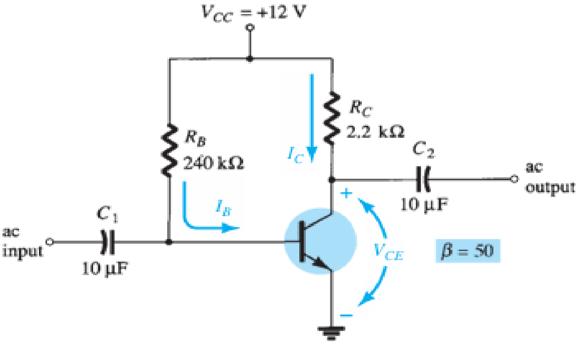
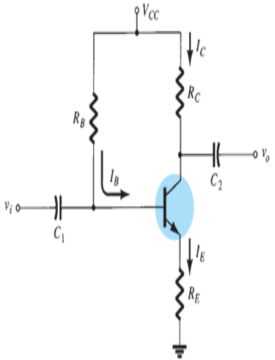


Figure 17: DC fixed-bias circuit for Example

Emitter-Bias Configuration

- The dc bias network of Fig. 18 contains an emitter resistor to improve the stability level over that of the fixed-bias configuration.
- The more stable a configuration, the less its response will change due to undesireable changes in temperature and parameter variations.



Base-Emitter Loop

- Writing Kirchhoff's voltage law around the indicated loop in the clockwise direction results in the following equation:
  $+V_{CC} - I_B R_B - V_{BE} - I_E R_E = 0$

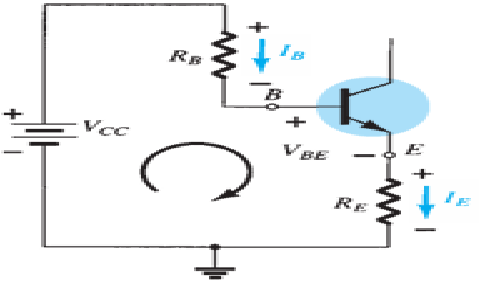


Figure 19: BJT bias circuit with emitter resistor

We know, $I_E = (\beta + 1)I_B$

$+V_{CC} - I_B R_B - V_{BE} - (\beta + 1)I_B R_E = 0$

$-I_B(R_B + (\beta + 1)R_E) + V_{CC} - V_{BE} = 0$

or $I_B(R_B + (\beta + 1)R_E) - V_{CC} + V_{BE} = 0$

$I_B(R_B + (\beta + 1)R_E) = V_{CC} - V_{BE}$

$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E}$

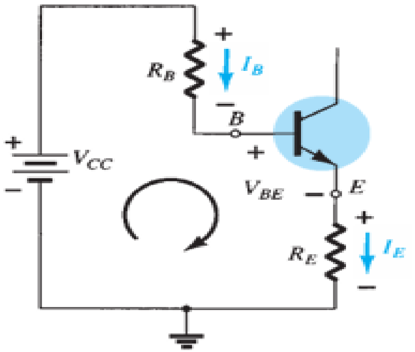
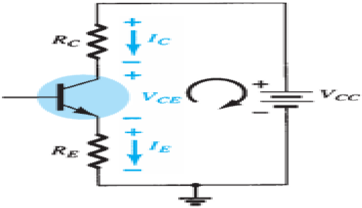


Figure 20

- Writing Kirchhoff's voltage law around the indicated loop in the clockwise direction results in the following equation:
  $+I_E R_E + V_{CE} + I_C R_C - V_{CC} = 0$
  Substituting $I_E \approx I_C$,
  $V_{CE} = V_{CC} + I_C(R_C + R_E)$
  $V_E = I_E R_E$
  $V_{CE} = V_C - V_E$
  $V_C = V_{CC} - I_C R_C$
  $V_B = V_{CC} - I_B R_B$
  or $V_B = V_{BE} + V_E$



Example

- Determine the following for the fixed-bias configuration:

(a) I_B and I_C
(b) V_{CE}
(c) V_B, V_E and V_C
(d) V_{BC}

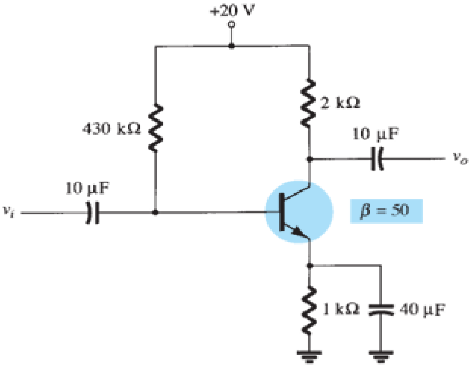
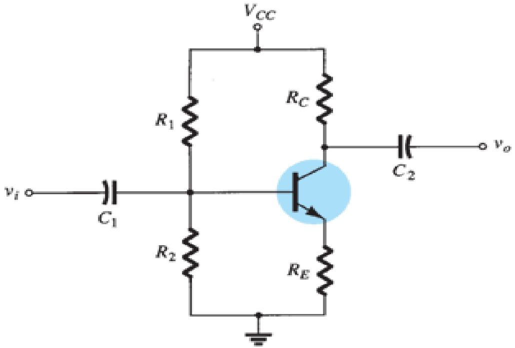


Figure 22: DC emitter-bias circuit for Example

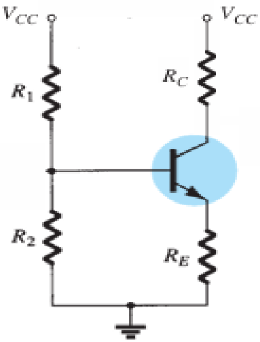
Voltage Divider-Bias Configuration

- In the previous bias configurations the bias current I_{CQ} and voltage V_{CE_Q} were a function of the current gain β of the transistor.
- However, because β is temperature sensitive, especially for silicon transistors, it would be desirable to develop a bias circuit that is less dependent on, or in fact is independent of, the transistor beta.
- The voltage-divider bias configuration of Fig. 23 is such a network.



Exact Analysis

- For the dc analysis the network of Fig. 23 can be redrawn as shown in Fig. 24.
- The input side of the network can then be redrawn as shown in Fig. 24 for the dc analysis.



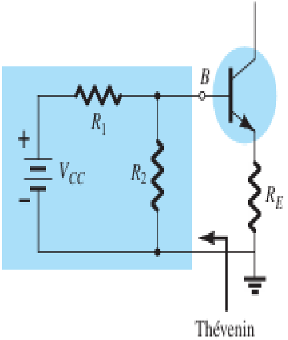


Figure 24

Exact Analysis

- The Thévenin equivalent network for the network to the left of the base terminal can then be found in the following manner:
  R_{Th}: The voltage source is replaced by a short-circuit equivalent as shown in Fig. 25:
  $R_{Th} = R_1 || R_2$

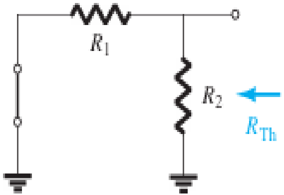


Figure 25: Determining R_{Th}

Exact Analysis

- E_{Th}: The voltage source V_{CC} is returned to the network and the open-circuit Thévenin voltage of Fig. 26 determined as follows:
- Applying the voltage-divider rule gives-
  $E_{Th} = V_{R_2} = \frac{R_2 V_{CC}}{R_1+R_2}$

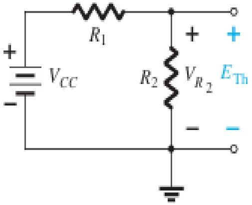


Figure 26: Determining E_{Th}

Exact Analysis

- The Thévenin network is then redrawn as shown in Fig. 27, and I_{B_Q} can be determined by first applying Kirchhoff's voltage law in the clockwise direction for the loop indicated:
  $E_{Th} - I_B R_{Th} - V_{BE} - I_E R_E = 0$
  Substituting $I_E = (\beta + 1) I_B$ and solving for I_B yields,
  $I_B = \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1) R_E}$

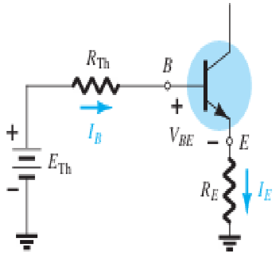


Figure 27: Inserting the Thévenin equivalent circuit.

Example

- Determine the following for the fixed-bias configuration:

(a) I_C

(b) V_{CE}

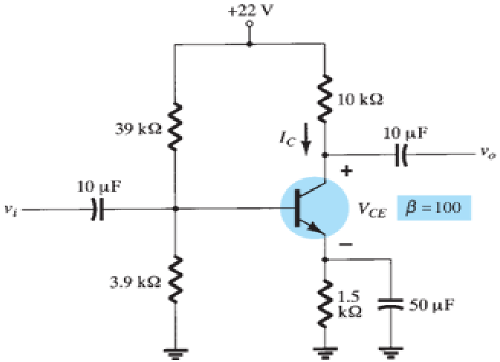


Figure 28: DC voltage divider-bias circuit for Example

Approximate Analysis

- The input section of the voltage-divider configuration can be represented by the network of Fig. 29.
- The resistance R_i is the equivalent resistance between base and ground for the transistor with an emitter resistor R_E.
- The reflected resistance between base and emitter is defined by $R_i = (\beta + 1)R_E$.

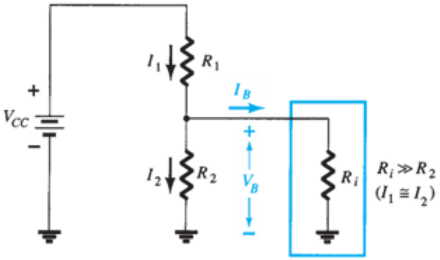


Figure 29

Approximate Analysis

- If R_i is much larger than the resistance R_2, the current I_B will be much smaller than I_2 (current always seeks the path of least resistance) and I_2 will be approximately equal to I_1.
- If we accept the approximation that I_B is essentially 0 A compared to I_1 or I_2, then $I_1 = I_2$, and R_1 and R_2 can be considered series elements.
- The voltage across R_2, which is actually the base voltage, can be determined using the voltage-divider rule (hence the name for the configuration).
  $V_B = \frac{R_2 V_{CC}}{R_1 + R_2}$

Approximate Analysis

- Because $R_i = (\beta+1)R_E \approx \beta R_E$, the condition that will define whether the approximate approach can be applied is
  $\beta R_E \geq 10R_2$
- Once V_B is determined, the level of V_E can be calculated from
  $V_E = V_B - V_{BE}$
- and the emitter current can be determined from
  $I_E = \frac{V_E}{R_E}$
  and
  $I_{C_Q} \approx I_E$
- The collector to emitter voltage is determined by
  $V_{CE} = V_{CC} - I_C R_C - I_E R_E$
  but because $I_E \approx I_C$
  $V_{CE_Q} = V_{CC} - I_C(R_C + R_E)$
- Since β does not appear in the above equations and I_B was not calculated. The Q -point (as determined by I_{CQ} and V_{CE_Q}) is therefore independent of the value of β.

- Determine V_C and V_B for the networks of Fig. 30.

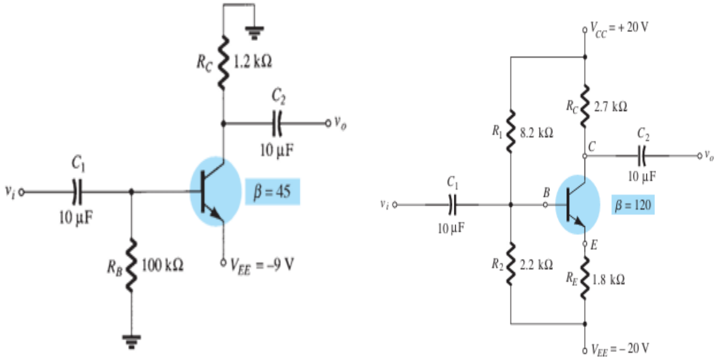


Figure 30

Bias Stabilization

- The stability of a system is a measure of the sensitivity of a network to variations in its parameters.
- In any amplifier employing a transistor the collector current I_C is sensitive to each of the following parameters:
  • β : increases with increase in temperature
  • V_{BE} : decreases about 2.5 mV per degree Celsius (°C) increase in temperature
  • I_{CO} (reverse saturation current): doubles in value for every 10°C increase in temperature.
- Any or all of these factors can cause the bias point to drift from the designed point of operation.

Bias Stabilization

- A stability factor S is defined for each of the parameters affecting bias stability as follows:
  $S(I_{CO}) = \frac{\Delta I_C}{\Delta I_{CO}}$
  $S(V_{BE}) = \frac{\Delta I_C}{\Delta V_{BE}}$
  $S(\beta) = \frac{\Delta I_C}{\Delta \beta}$
- Networks that are quite stable and relatively insensitive to temperature variations have low stability factors.
- The higher the stability factor, the more sensitive is the network to variations in that parameter.

$S(I_{CO})$

- Foe Fixed-Bias Configuration : $S(I_{CO}) \approx \beta$
- For Emitter-Bias Configuration : $S(I_{CO}) \approx \frac{\beta(1+\frac{R_B}{R_E})}{\beta+\frac{R_B}{R_E}}$

For $\frac{R_B}{R_E} >> \beta, S(I_{CO}) \approx \beta$
For $\frac{R_B}{R_E} << 1, S(I_{CO}) \approx 1$
For $1 << \frac{R_B}{R_E} << \beta + 1, S(I_{CO}) \approx \frac{R_B}{R_E}$

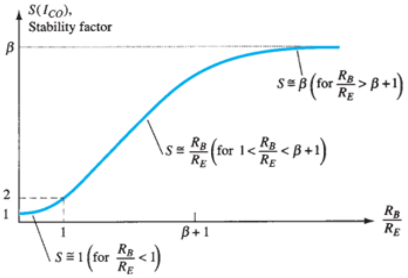


Figure 31: Variation of stability factor $S(I_C O)$ with the resistor ratio $\frac{R_B}{R_E}$ for the

$S(I_{CO})$

- For Voltage Divider Bias:

$$S(I_{CO}) = \frac{\beta(1+\frac{R_B}{R_C})}{\beta+\frac{R_B}{R_C}}$$

$S(V_{BE})$

- For the fixed-bias configuration: $S(V_{BE}) \approx \frac{-\beta}{R_B}$
- For the emitter-bias configuration: $S(V_{BE}) \approx \frac{\frac{-\beta}{R_E}}{\beta + \frac{R_B}{R_E}}$

  Substituting the condition $\beta >> \frac{R_B}{R_E}$ results in the following equation for $S(V_{BE})$: $S(V_{BE}) \approx \frac{\frac{-\beta}{R_E}}{\beta} = -\frac{1}{R_E}$
- For Voltage Divider Bias: $S(V_{BE}) = \frac{\frac{-\beta}{R_E}}{\beta + \frac{R_{Th}}{R_E}}$

$S(\beta)$

- For the fixed-bias configuration: $S(\beta) = \frac{I_{C_1}}{\beta_1}$
- For the emitter-bias configuration: $S(\beta) = \frac{\Delta I_C}{\Delta \beta} = \frac{I_{C_1}(1+\frac{R_B}{R_E})}{\beta_1(\beta_2+\frac{R_B}{R_E})}$
- For Voltage Divider Bias: $S(\beta) = \frac{\Delta I_C}{\Delta \beta} = \frac{I_{C_1}(1+\frac{R_{Th}}{R_E})}{\beta_1(\beta_2+\frac{R_{Th}}{R_E})}$

Small Signal Analysis(h parameter model) of CE configuration

- The description of the hybrid equivalent model will begin with the general two-port system of Fig. 32.

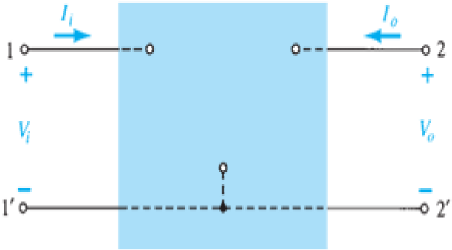


Figure 32: Two port network.

- The following set of equations is only one of a number of ways in which the four variables of Fig. 32 can be related.
  $V_i = h_{11} I_i + h_{12} V_o$
  $I_o = h_{21} I_i + h_{22} V_o$
- The parameters relating the four variables are called h-parameters , from the word "hybrid."
- The term hybrid was chosen because the mixture of variables (V and I) in each equation results in a "hybrid" set of units of measurement for the h-parameters.

- h_{11}: If we arbitrarily set $V_o = 0$ (short circuit the output terminals) and solve for h_{11} in first equation in the previous slide, we find

  $h_{11} = \frac{V_i}{I_i}\Big|_{V_o=0}$ ohms

- The ratio indicates that the parameter h_{11} is an impedance parameter with the units of ohms.
- Because it is the ratio of the input voltage to the input current with the output terminals shorted, it is called the short-circuit input-impedance parameter.
- The subscript 11 of h_{11} refers to the fact that the parameter is determined by a ratio of quantities measured at the input terminals.

h_{12}

- h_{12}: If I_i is set equal to zero by opening the input leads, the following results for h_{12}:

$$h_{12} = \left.\frac{V_i}{V_o}\right|_{I_i=0} \text{ unitless}$$

- The parameter h_{12}, therefore, is the ratio of the input voltage to the output voltage with the input current equal to zero.
- It has no units because it is a ratio of voltage levels and is called the open-circuit reverse transfer voltage ratio parameter.
- The subscript 12 of h_{12} indicates that the parameter is a transfer quantity determined by a ratio of input (1) to output (2) measurements.

h_{21}

- h_{21}: If in second equation, V_o is set equal to zero by again shorting the output terminals, the following results for h_{21}:

  $h_{21} = \frac{I_o}{I_i}\Big|_{V_o=0}$ unitless

- The parameter h_{21} is the ratio of the output current to the input current with the output terminals shorted.
- This parameter, like h12 , has no units because it is the ratio of current levels.
- It is formally called the short circuit forward transfer current ratio parameter.

h_{22}

- h_{22}: The last parameter, h_{22}, can be found by again opening the input leads to set $I_1 = 0$ and solving for h_{22} in the second equation:

$$h_{22} = \frac{I_o}{V_o}\bigg|_{I_i=0} \text{ siemens}$$

- Because it is the ratio of the output current to the output voltage, it is the output conductance parameter, and it is measured in siemens (S).
- It is called the open-circuit output admittance parameter.
- The subscript 22 indicates that it is determined by a ratio of output quantities.

- Because each term of the first equation has the unit volt, let us apply Kirchhoff's voltage law "in reverse" to find a circuit of Fig. 33 that "fits" the equation.

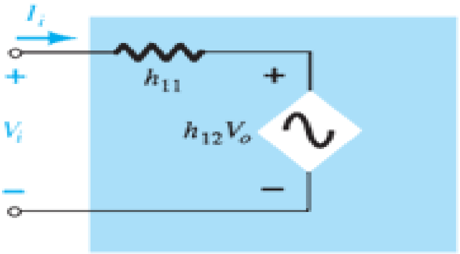


Figure 33: Hybrid input equivalent circuit.

- Because each term of the second equation has the units of current, let us now apply Kirchhoff's current law "in reverse" to obtain the circuit of Fig. 34.

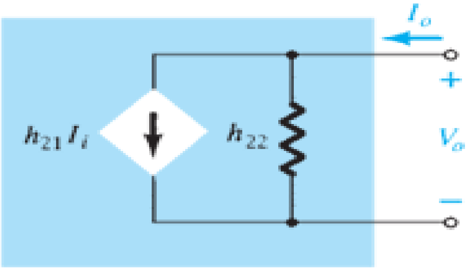


Figure 34: Hybrid output equivalent circuit.

- The complete "ac" equivalent circuit for the basic three-terminal linear device is indicated in Fig. 35 with a new set of subscripts for the h-parameters.
  h_{11} -> input resistance -> h_i
  h_{12} -> reverse transfer voltage ratio -> h_r
  h_{21} -> forward transfer current ratio -> h_f
  $h22$ -> output conductance -> h_o

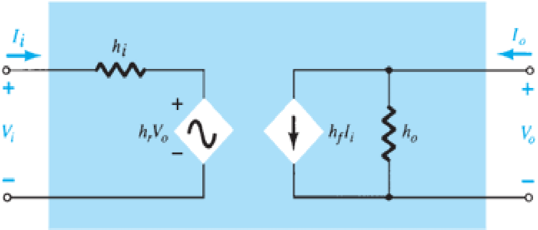


Figure 35: Hybrid output equivalent circuit.

CE h-parameter model

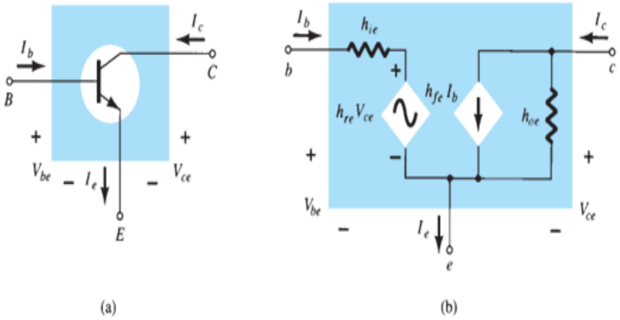


Figure 36: Hybrid output equivalent circuit.

- h parameters are real numbers at audio frequencies(20-20kHz), are easy to measure, can also be obtained from the transistor static characteristics curves, and are particularly convenient to use in circuit analysis and design.

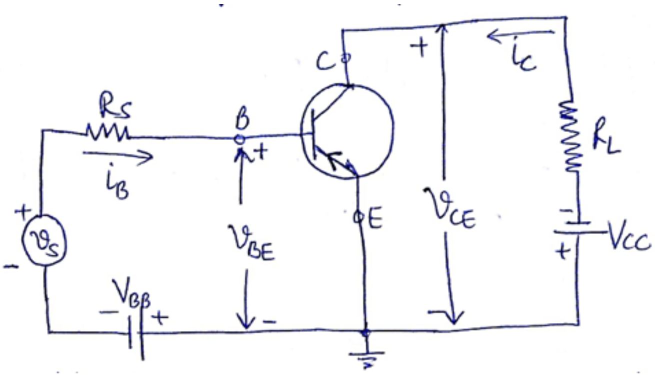


Figure 37

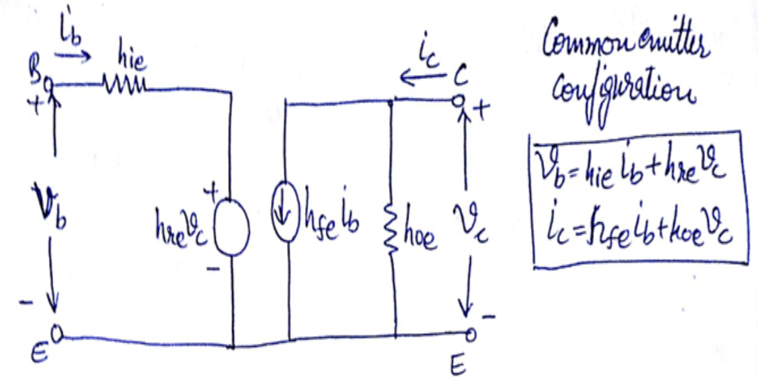


Figure 38: CE h-parameter model

Typical h-parameter values for a CE transistor at $I_E =$ 1.3mA

$h_{11} = h_{ie} = 1100\Omega$
$h_{12} = h_{re} = 2.5x10^{-4}$
$h_{21} = h_{fe} = 50$
$h_{22} = h_{oe} = 24\mu A/V$
$1/h_o = 40K$

Analysis of a CE transistor amplifier circuit using h-parameters

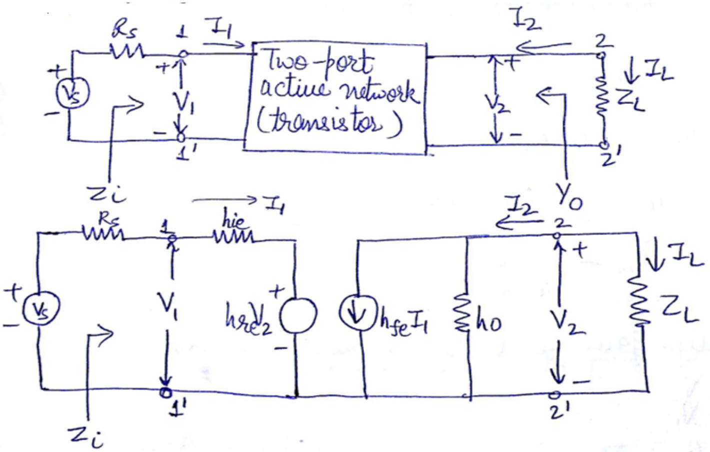


Figure 39

Analysis of a CE transistor amplifier circuit using h-parameters

- **The current gain or current amplification,** A_I

$A_I = \frac{I_L}{I_1} = -\frac{I2}{I1}$

Since $I_2 = h_{fe} I_1 + h_{oe} V_2$ and $V_2 = I_L Z_L = -I_2 Z_L$

$A_I = -\frac{I_2}{I_1} = \frac{-h_{fe}}{1+h_{oe} Z_L}$

Analysis of a CE transistor amplifier circuit using h-parameters

- **The input impedence,** Z_i

  The impedence we see looking into the amplifier input terminals (1, 1') is the amplifier input impedence Z_i.

  $Z_i = \frac{V_1}{I_1}$

  From the input circuit, we have $V_1 = h_{ie} I_1 + h_{re} V_2$
  Therefore, $Z_i = \frac{h_{ie} I_1 + h_{re} V_2}{I_1} = h_{ie} + h_{re} \frac{V_2}{I_1}$

  Since $V_2 = -I_2 Z_L = A_I I_1 Z_L$

  $Z_i = h_{ie} + h_{re} A_I Z_L$
  Also $Y_L = 1/Z_L$ and $A_I = \frac{-hfe}{1+h_{oe} Z_L}$
  Therefore, $Z_i = h_{ie} - \frac{h_{fe} h_{re}}{Y_L + h_{oe}}$

Analysis of a CE transistor amplifier circuit using h-parameters

- **The voltage gain or voltage amplification,** A_V
  $A_V = \frac{V_2}{V_1}$

  $V_2 = A_I I_1 Z_L$

  $A_V = \frac{A_I Z_L}{Z_i}$

  $A_V = \frac{A_I I_1 Z_L}{V_1} = \frac{A_I Z_L}{Z_i}$

Analysis of a CE transistor amplifier circuit using h-parameters

- **The output admittance,** Y_O
- The output impedence $Z_O = 1/Y_O$ is obtained by setting the source voltage V_S to zero and the load impedence Z_L to infinity and by driving the output terminals from generator V_2. If the current drawn from V_2 is I_2 then,

  $Y_O = \frac{I_2}{V_2}$ with V_S=0 and $R_L = Z_L = \infty$

  $I_2 = h_{fe} I_1 + h_{oe} V_2$

  $\frac{I_2}{V_2} = h_{fe} \frac{I_1}{V_2} + hoe \frac{V_2}{V_2}$

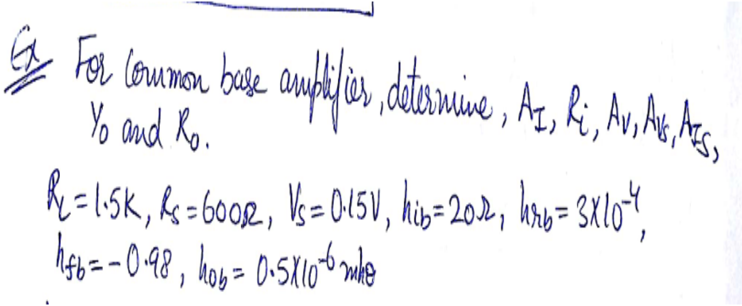
  $Y_O = h_{fe} \frac{I_1}{V_2} + h_{oe}$

  With $V_S = 0$, in figure, we have,

  $R_S I_1 + h_{ie} I_1 + h_{re} V_2 = 0$

  $\frac{I_1}{V_2} = \frac{-h_{re}}{h_{ie} + R_S}$

  Therefore, $Y_O = h_{oe} - \frac{h_{fe} h_{re}}{h_{ie} + R_S}$



Ex For Common base amplifier, determine, A_I, R_i, A_V, A_{Vs}, A_{Is}, Y_o and R_o.

$R_L = 1.5K$, $R_s = 600\Omega$, $V_s = 0.15V$, $h_{ib} = 20\Omega$, $h_{rb} = 3\times10^{-4}$, $h_{fb} = -0.98$, $h_{ob} = 0.5\times10^{-6}$ mho

Figure 40

END OF MODULE 2